GalaxyGAN: Conditional Generation of Galaxy Morphologies with WGAN-GP

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The project will use the **Galaxy10 DECals Dataset** from AstroNN (https://astronn.readthedocs.io/en/latest/galaxy10.html). This dataset consists of about 17,700 galaxy images labeled with 10 morphology categories at 69×69 resolution. There exist high-resolution alternatives to this dataset at 256×256 on HuggingFace (https://huggingface.co/datasets/matthieulel/galaxy10_decals). The choice between low and high resolutions remains to be discussed with the TA and professor depending on available computational power via feedback of this proposal.

I found this dataset from the **Dark Energy Camera Legacy Survey (DECaLS)**. It includes 10 distinct galaxy morphology classes, such as spiral, elliptical, edge-on disk, and merging galaxies.

This dataset is well-suited for generative modeling due to:

- Its scientifically meaningful class labels, which enable structured conditional generation
- Its moderate size (the 69x69 resolution dataset most likely), allowing training from scratch on a single GPU
- The diversity of visual structures, making it ideal for evaluating model expressiveness
- ~17,700 images
- Source: DECaLS (public astronomical survey)
- Labels: 10 galaxy morphology classes

Model Architecture(s)

I propose to use a Conditional GAN with a generator inspired by DCGAN but with a WGAN-GP loss function. It will generate images by receiving a concatenation of a 128-dimensional noise vector and a class-embedding vector corresponding to galaxy morphology.

For the discriminator, the combination of a real/fake image and a class-embeddings vector will be passed. Then, convolutional layers will be used to discriminate between images. Finally, the Wasserstein loss with a gradient penalty will be implemented to stabilize training and avoid mode collapse.

In this setup, class-conditioned images can be generated for any of the 10 galaxy morphologies. Also, the performance difference between cGAN and unconditional GAN will be evaluated to measure the impact of conditioning on the generation quality.

Extra Criteria

To make this project above average, the following criteria will be met:

1. Conditional Generation by Morphology

Generation of class-conditioned images according to the galaxy morphological structure (spiral, elliptical, merging, etc.) will be supported. Conditioning will be done through label embeddings used both by generator and discriminator.

2. Latent Space Exploration

- Perform latent interpolation within the same morphology class to visualize smooth transitions between generated galaxy structures
- Use t-SNE or PCA to visualize latent representations and analyze clustering across morphology classes

3. Metrics & Training Evaluation

- Track changes in Fréchet Inception Distance (FID) and Inception Score during the training process
- Evaluate FID per each galaxy morphology type for comparing generation quality among them
- Log generator/discriminator losses, gradient norms, and generated samples using Weights & Biases

4. Interactive Gallery (GUI)

Implementing Gradio will enable to perform the following actions:

- Select a galaxy morphology class
- Generate galaxy images by specifying different random seeds
- Explore latent space interpolations between generated galaxies

I anticipate that the interface will provide an accessible and interactive demonstration of the model's conditional generation capabilities.